

Mini-Projet:

GL

2025/2026
L3:SCI

***the design of a national
emergency management
system that enables citizens to
report emergencies via a
mobile application or a phone
call.***

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Project Overview: Emergency Management System

1. Project Description

Overview of the system and its purpose.

2. Reporting Options

- Option 1 – Mobile Application
- Option 2 – Phone Call

3. Actors

Primary and secondary actors who interact with the system.

4. Functional Requirements

Key features and capabilities of the system.

5. Non-Functional Requirements

Performance, security, and usability constraints.

6. Descriptive Sheets

Detailed scenarios for:

- Authentication
- Report Emergency
- Process Emergency Call
- Alert Unit Center

7. UML Diagrams

- Use Case Diagram
- Class Diagrams (Entities & Controllers& Participant)
- Sequence Diagrams, detailed Sequence Diagrams

8. System Architecture (MVC Pattern)

- Model: Data management
- View: Interfaces (app, agent)
- Controller: Logic (authentication, emergency, alert, dispatch)

9. Wireframes

Screens for citizen app and agent interface.

1. Description of the Project

The emergency management system is designed to allow citizens to report emergencies and enable emergency services to respond quickly and efficiently. The system supports two main ways of reporting emergencies: through a mobile application or through a phone call to the emergency center.

Option 1 – Mobile Application

In this option, the citizen uses a mobile application to report an emergency. First, the citizen authenticates using valid credentials, which allows the system to identify the user. After authentication, the citizen selects the option “Report Emergency” and enters the required emergency information, such as the type of emergency and additional details.

The system automatically retrieves the citizen’s stored information and GPS location. The citizen may optionally add extra information such as a textual description, pictures, or voice messages.

Once the emergency report is submitted, the system validates the information. If the emergency is new, the system alerts the unit center. After receiving the alert, the emergency center dispatches the appropriate emergency units (ambulance, fire, or police).

Option 2 – Phone Call

In this option, the citizen reports an emergency by making a phone call to the emergency center. An emergency agent answers the call and authenticates to access the emergency system (agent interface). The agent then processes the emergency call by collecting essential information from the caller, such as the location, type of emergency, and other relevant details.

The agent enters the collected information into the emergency management system. The system validates and saves the emergency information. If the emergency is new,

the system alerts the emergency center. Based on the received alert, the emergency center dispatches the appropriate unit center.

Requirements

Analysis

Phase

1. Requirements Expression :

Specification Model

Actors

The actors who interact with the Emergency Management System are:

- **Primary Actors:**
 - Citizen
 - Emergency Agent
 - **Secondary Actors:**
 - Unit center
-

Functional Requirements

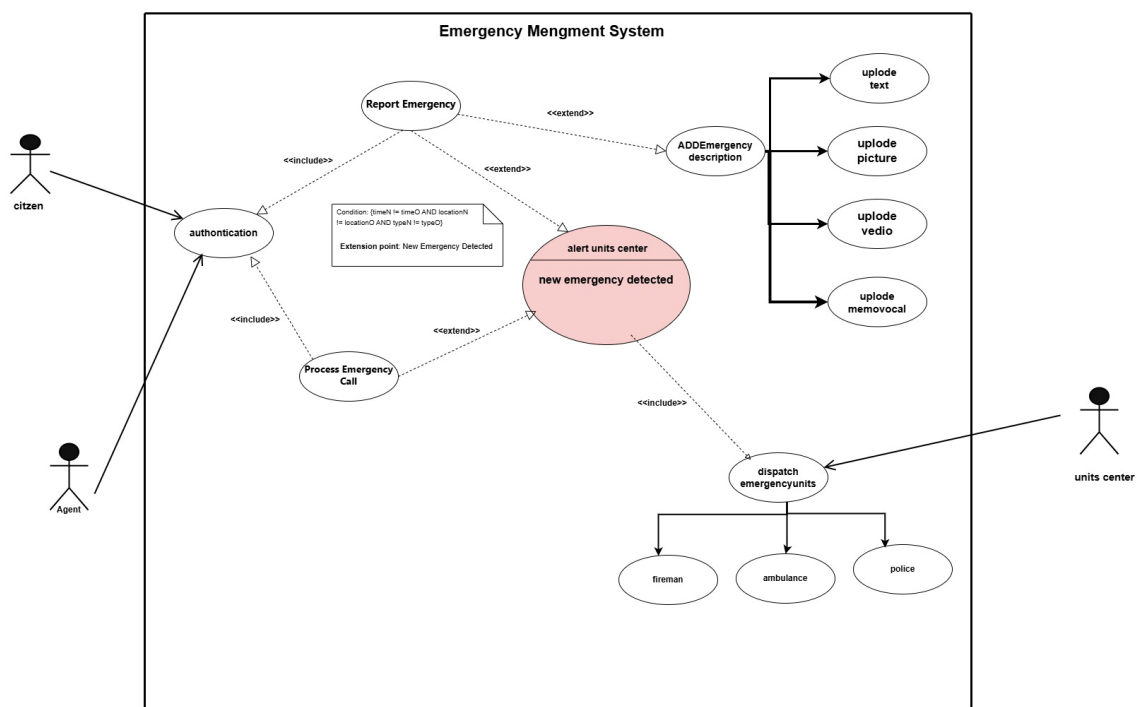
The Emergency Management System allows:

- User authentication using a phone number or credentials
 - Emergency reporting through a mobile application
 - Emergency reporting through a phone call
 - Automatic collection of emergency information after the user enters it
 - Optional addition of emergency description (text, pictures, voice messages)
 - Emergency call processing by an emergency agent
 - Validation and evaluation of emergency information
 - Detection of duplicate emergencies
 - Alerting the unit center when a new emergency is detected
 - Dispatching the appropriate emergency units
-

Non-Functional Requirements

- Authentication is required before reporting an emergency through the mobile application
- The system must be available 24/7
- The system must provide fast and reliable response time
- The system must ensure data accuracy, privacy, and security
- The user interface agent and the app must be simple and easy to use

Modilase the use case diagram



Descriptive Sheets

Descriptive Sheets: Authentication

NAME	Authentication
Actor	Citizen, Agent
Objective	The actor wants to securely access the emergency system.
Precondition	The actor possesses a valid username and password.
Normal Scenario	1. The actor opens the emergency system. 2. The system displays the authentication page. 3. The actor fills in the fields (username, password). 4. The system validates the credentials. 5. The system displays the main menu.
Alternative Scenario	3.1 Username and/or password incorrect. 3.1.1 The system displays an error message and returns to step 2.
Post-condition	the actor is successfully logged in and can access the emergency system.

Descriptive Sheets Report Emergency

NAME	Report Emergency
Actor	Citizen
Objective	The citizen wants to report an emergency so that the emergency center can respond
Precondition	The citizen is authenticated
Normal Scenario	1. The system prompts for emergency information. 2. The citizen enter the emergency information 3. The system save and validates the information.
Alternative Scenario	2.1 Missing or invalid information. 2.1.1 The system displays an error message and returns to step 2
Post-condition	Emergency is successfully reported and the citizen is notified

Descriptive Sheets Process Emergency Call

NAME	Process Emergency Call
Actor	Agent
Objective	the agent records and process incoming emergency call
Precondition	Agent is authenticated

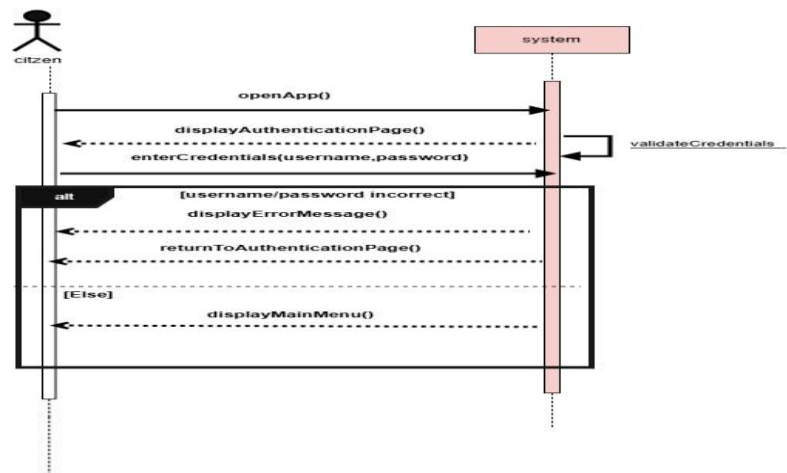
Normal Scenario	1. The agent receives an emergency call and authenticates 2. The agent collects emergency details from the caller 3. The system validates and saves the emergency information
Alternative Scenario	2.1 Caller provides unclear information 2.1.1 The agent asks follow-up questions and returns to step 2
Post-condition	the emergency call is processed and emergency information is stored in the system

Descriptive Sheets Alert unit Center

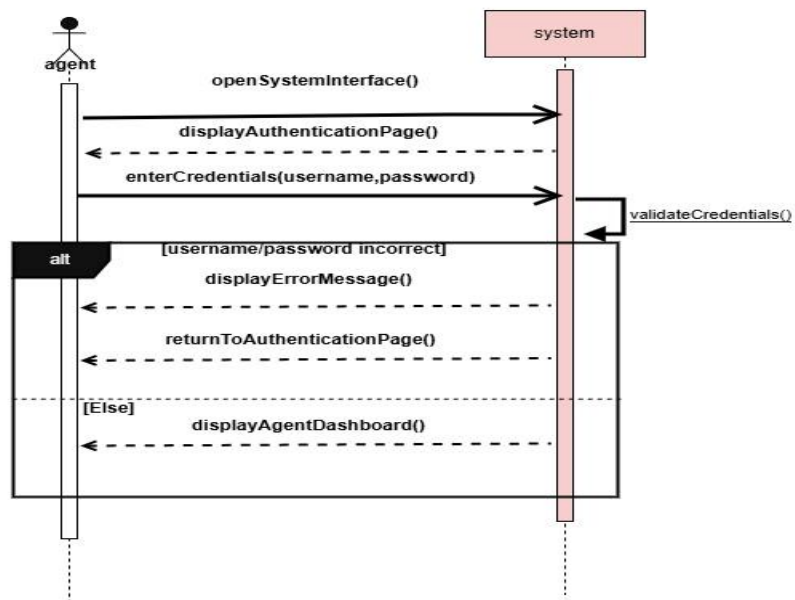
NAME	Alert unit Center
Actor	unit center
Objective	Notify the unit center about a new emergency
Precondition	A validated emergency report has been submitted; it is new (not duplicate)
Normal Scenario	1. System receives emergency report 2. System checks if the emergency is new 3. System sends alert to the unit center
Alternative Scenario	2.1 Emergency already reported 2.1.1 already reported message show up the system user (citizen, agent)
Post-condition	Unit center receives the alert

Sequence diagrams

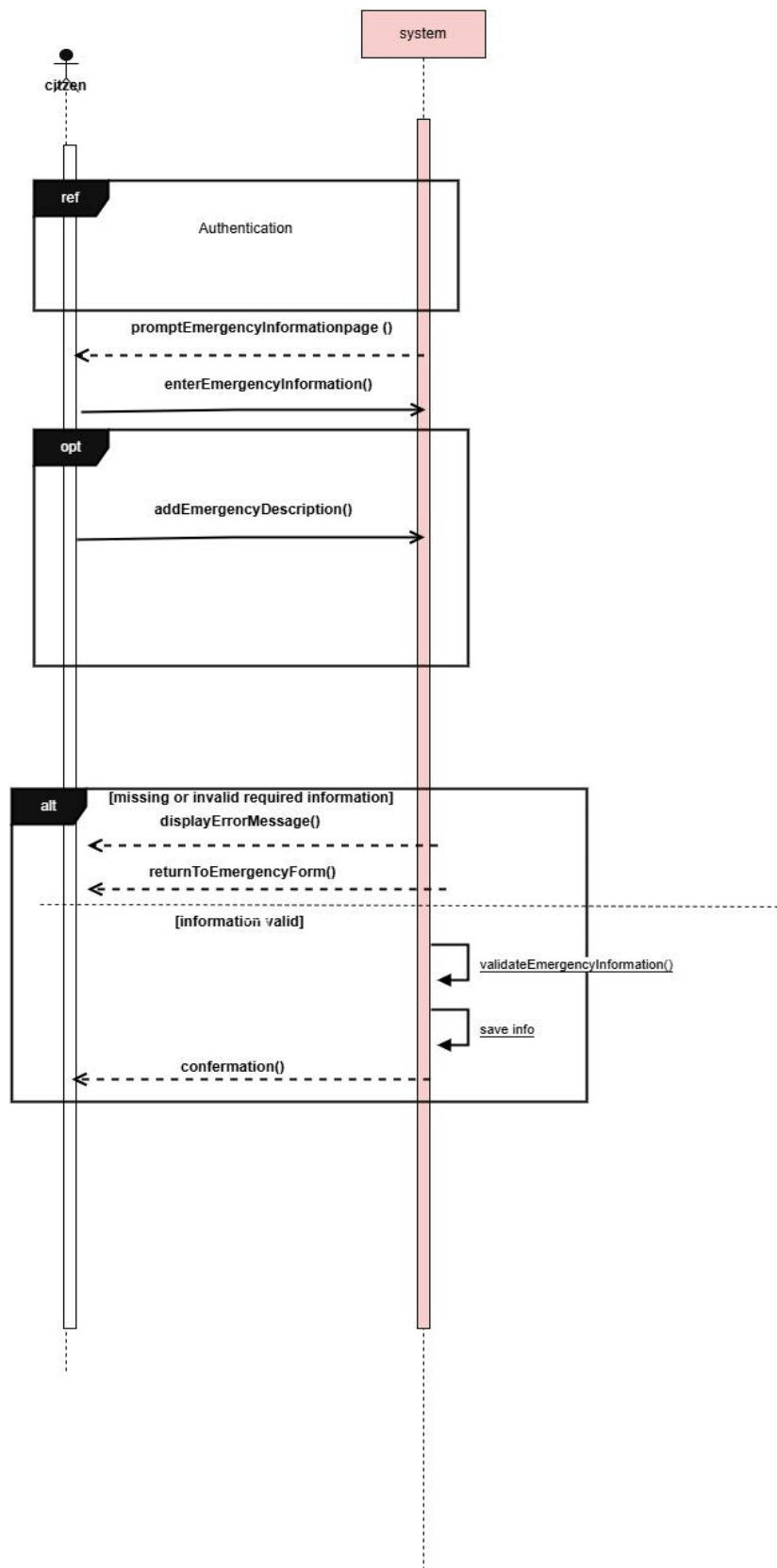
Authenticatation(app user)



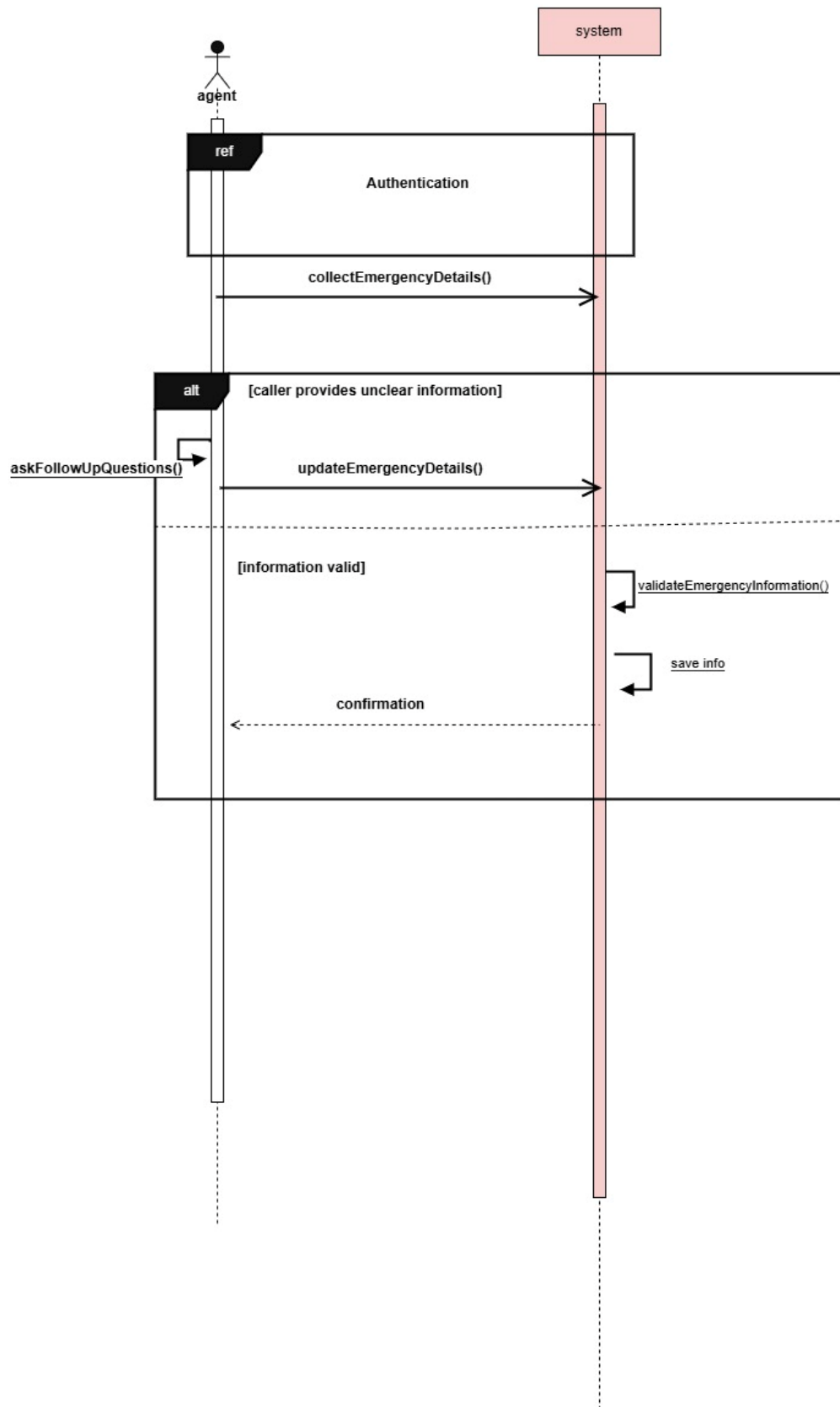
Authenticatation(agent interface)



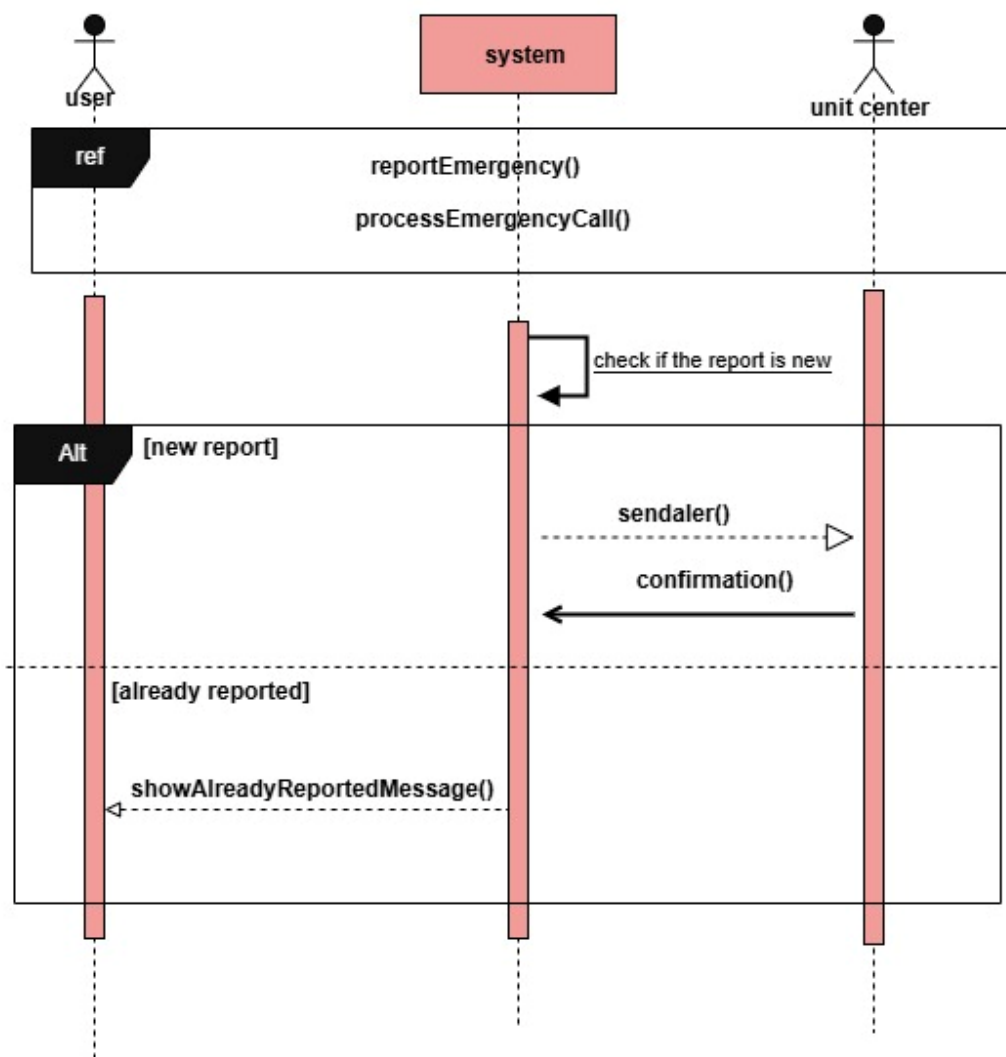
Report Emergency



Process Emergency Call

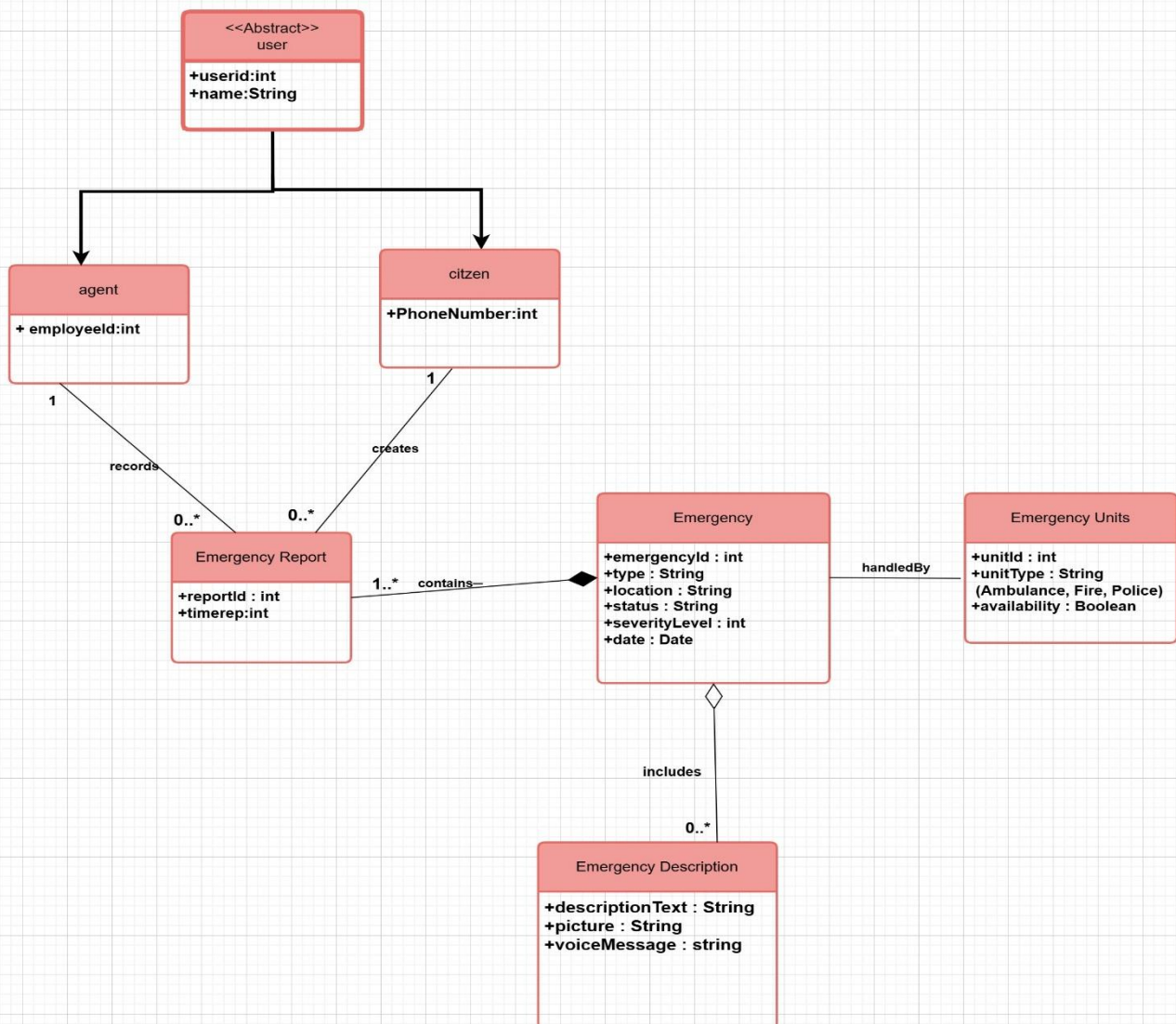


Alert unit Center



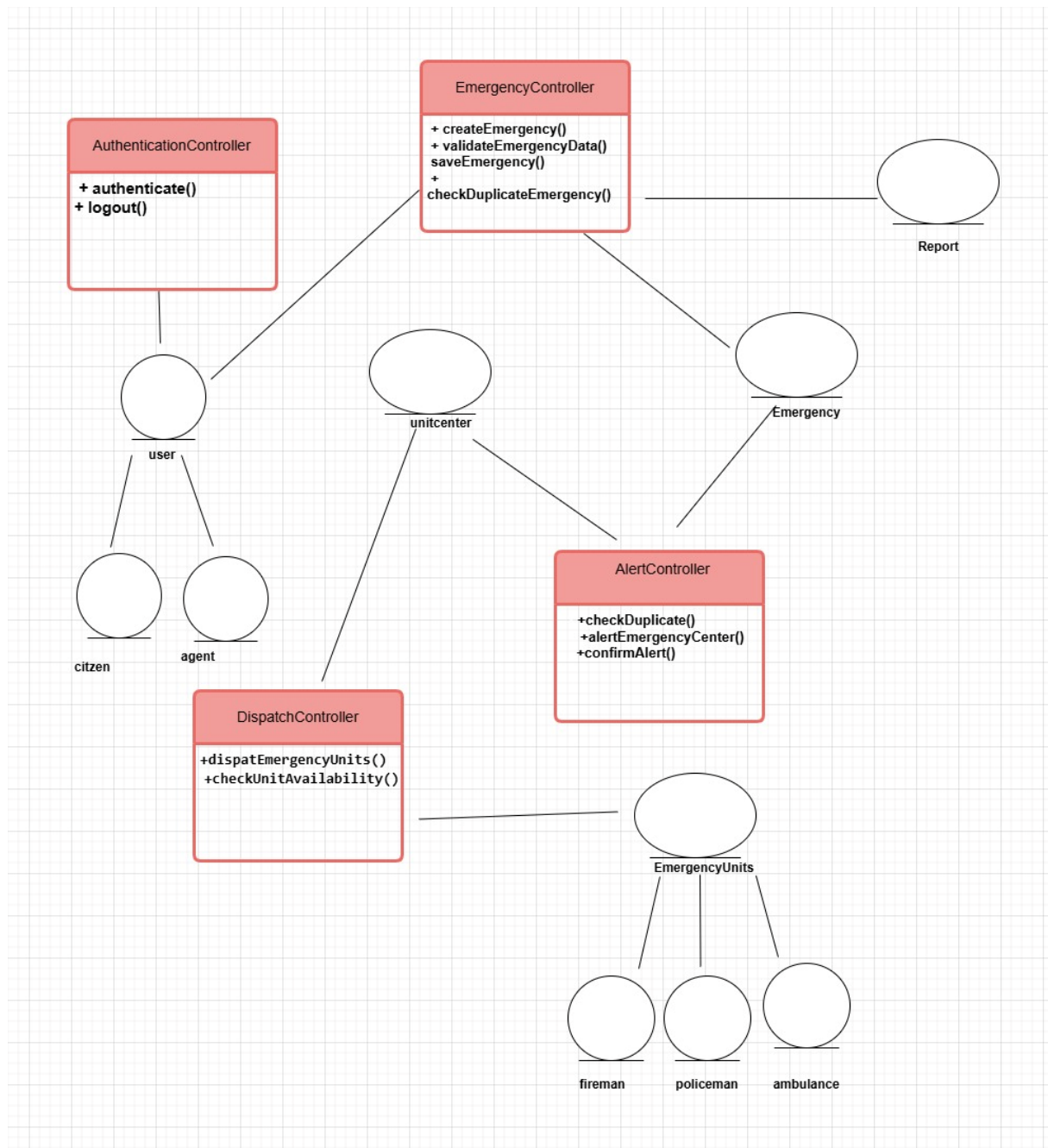
Phase *d'Analyse*

Diagram of Classes Entités

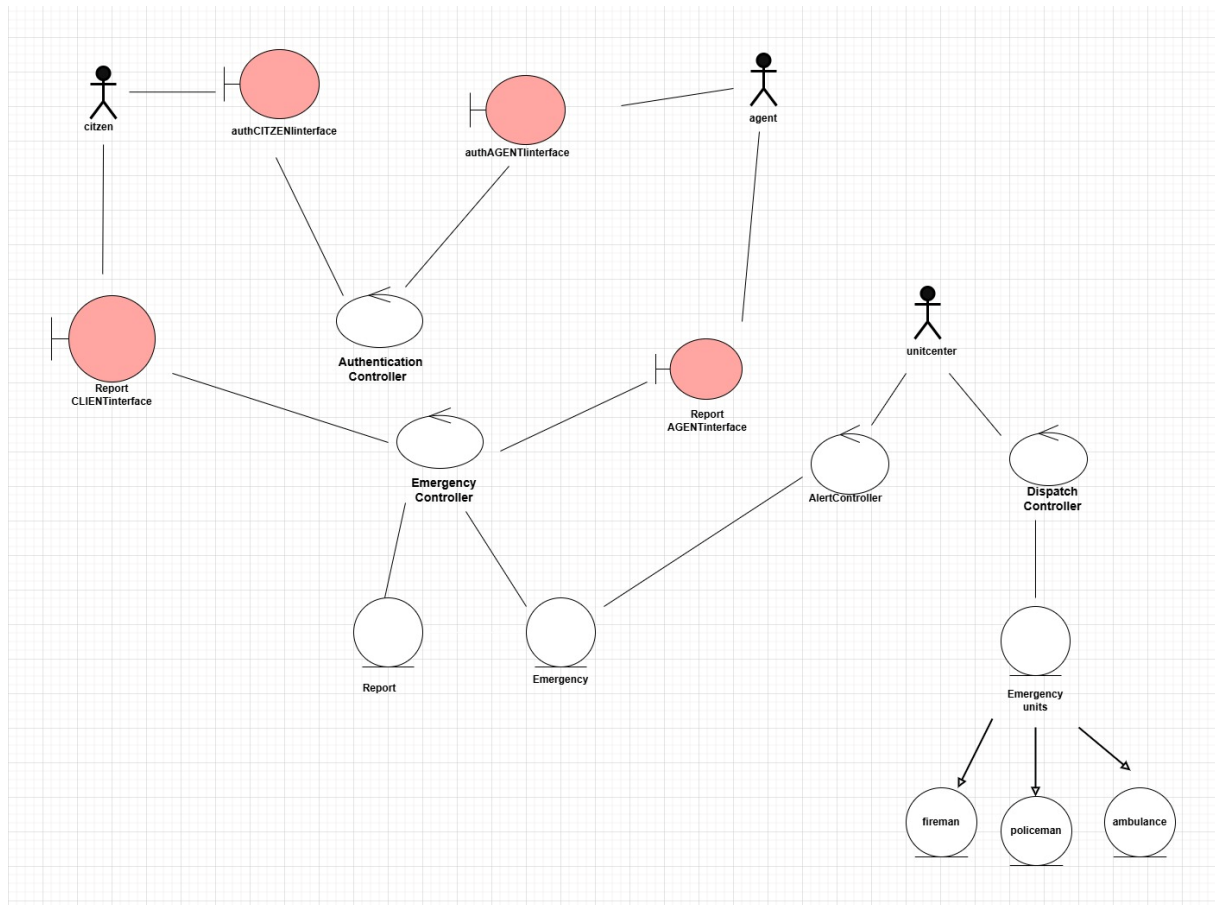


Phase de *Conception*

Diagram Classes controller

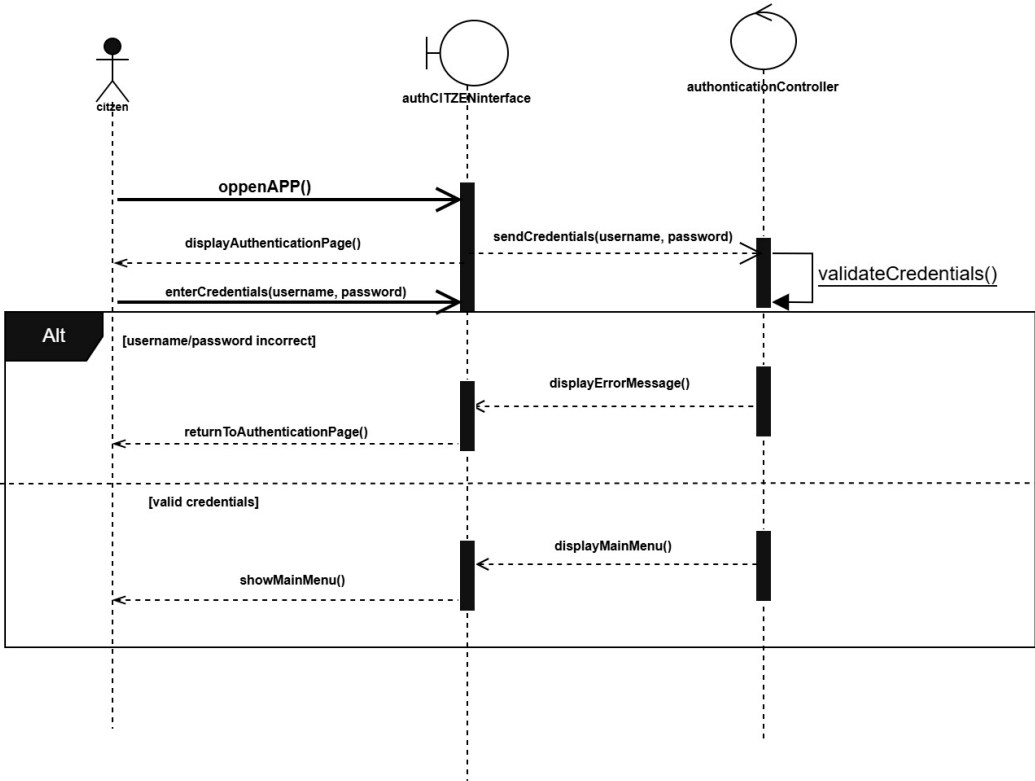


participant class diagram

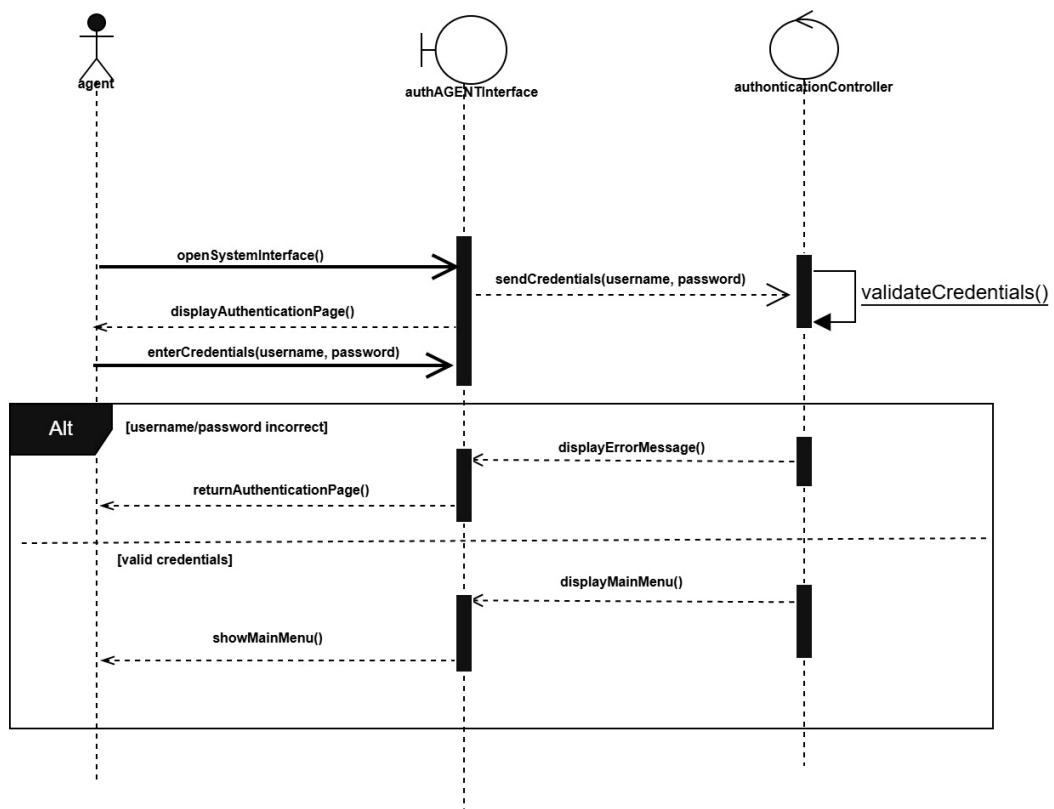


Detailed Sequence diagrams

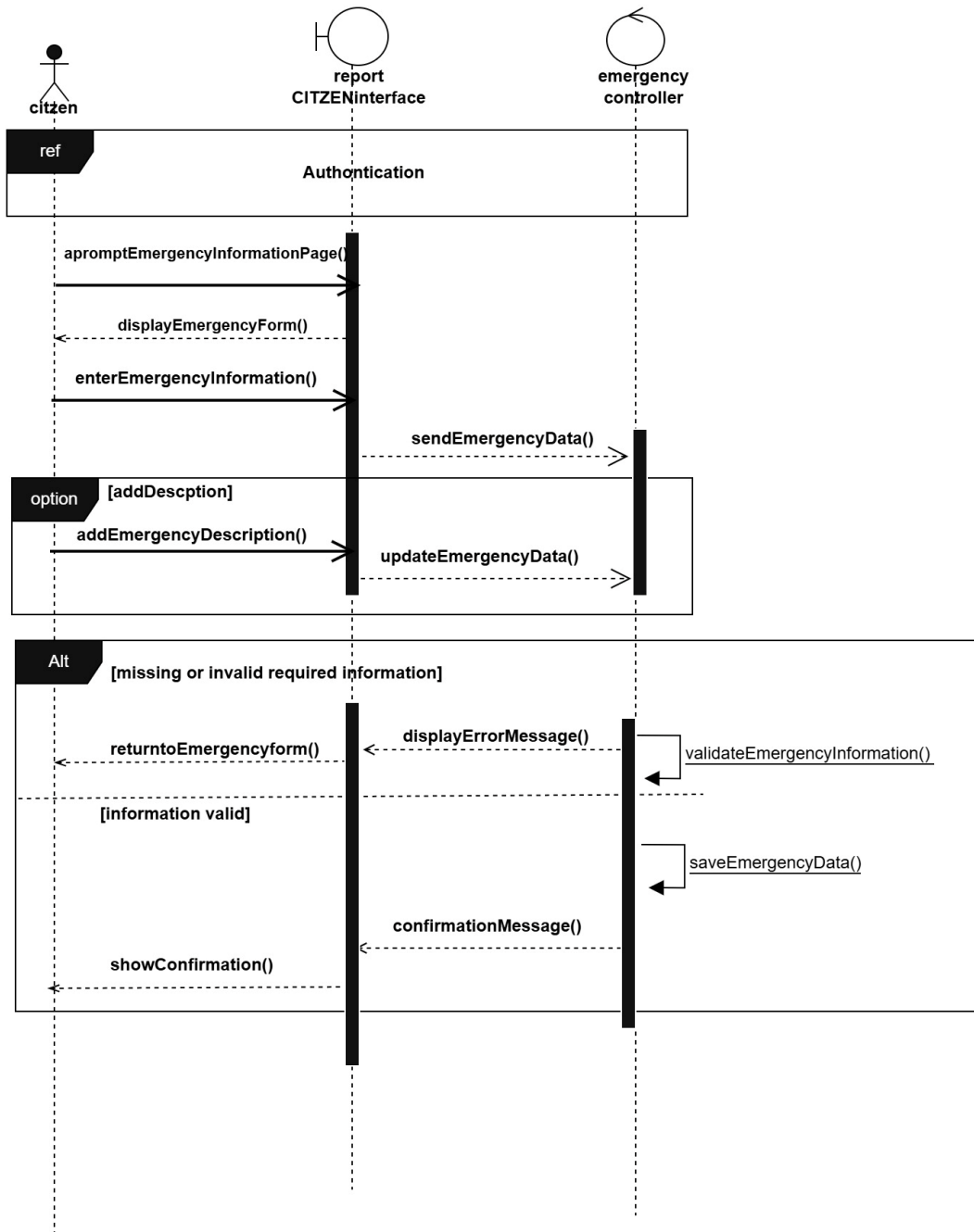
Authenticatation(app user)



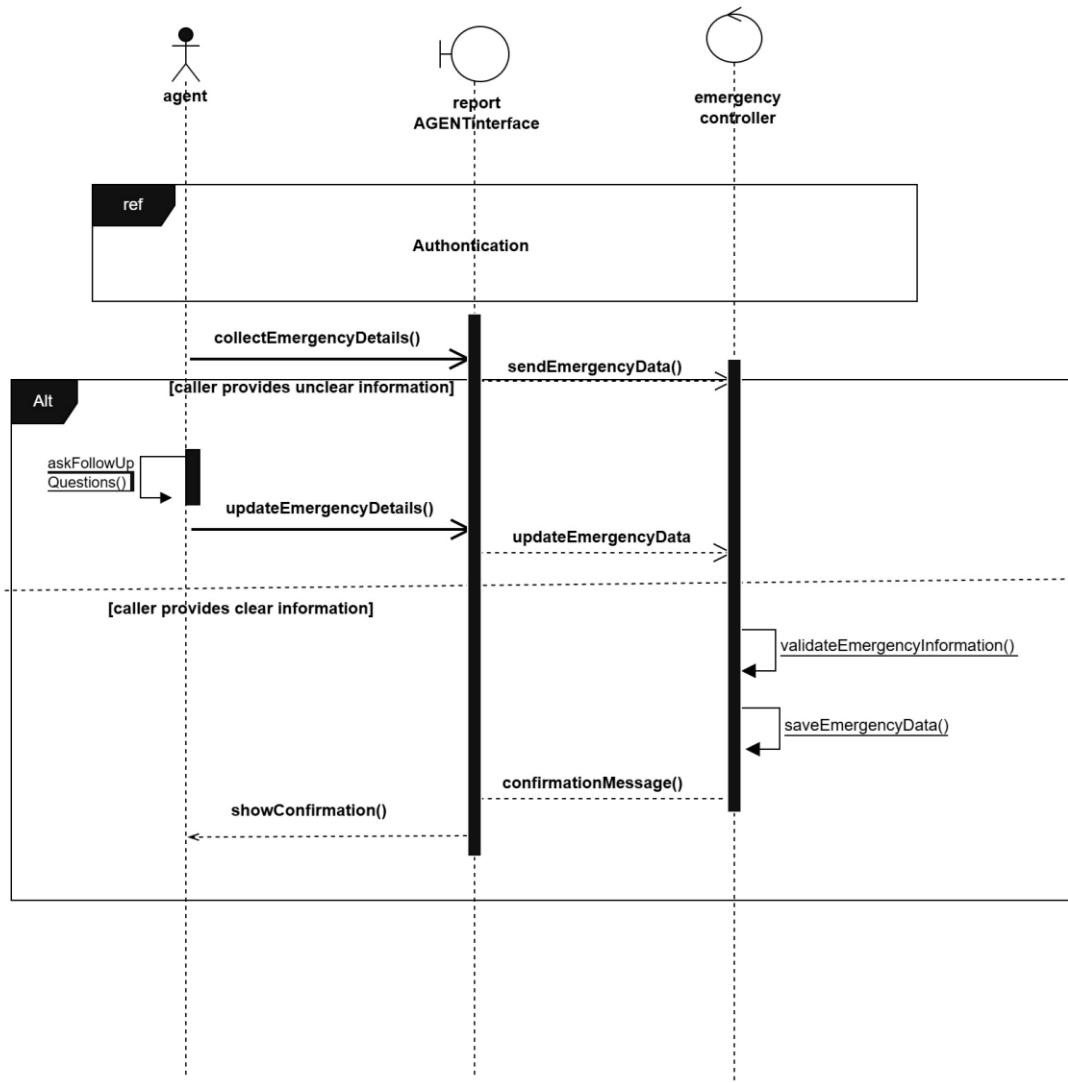
Authenticatation(agent interface)



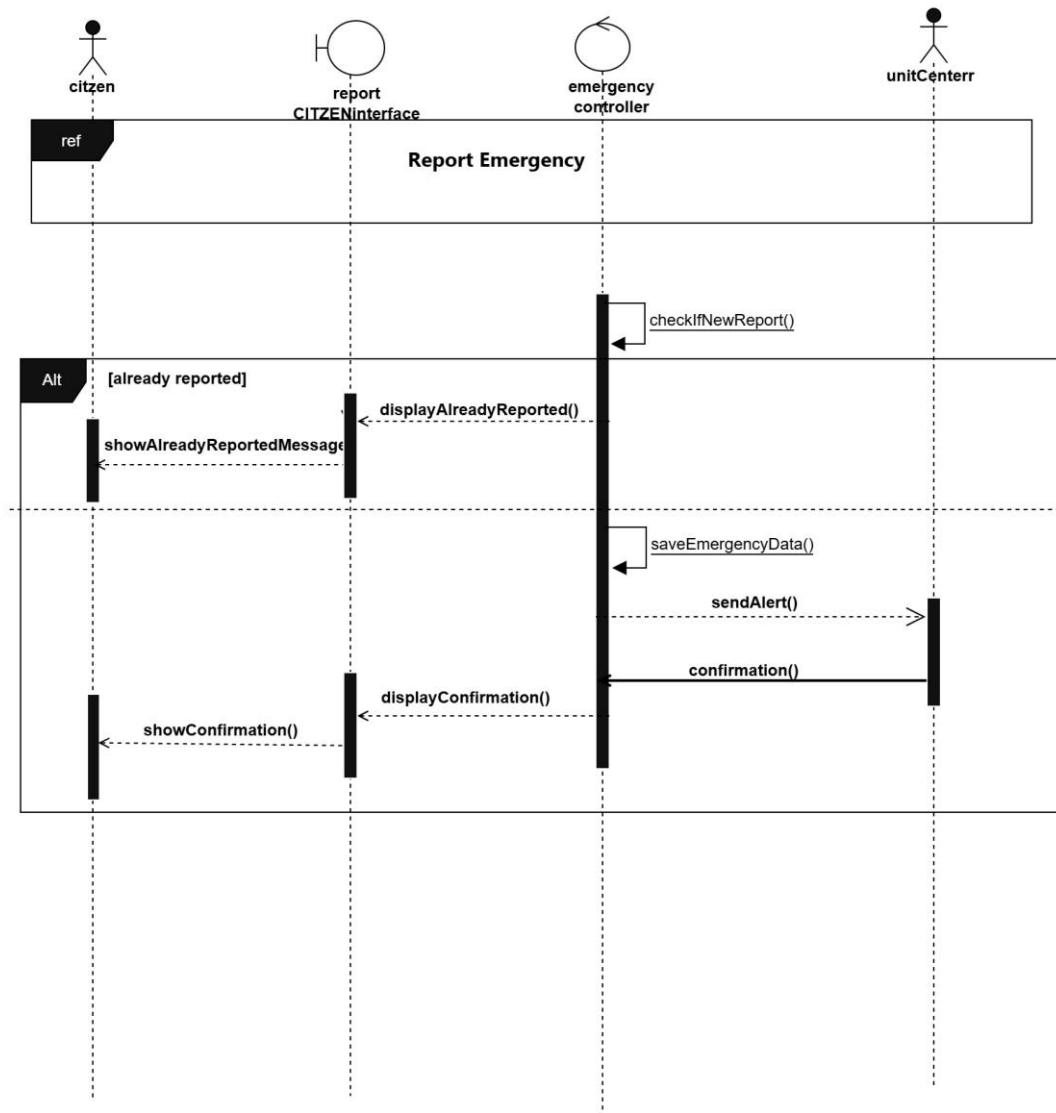
Report Emergency



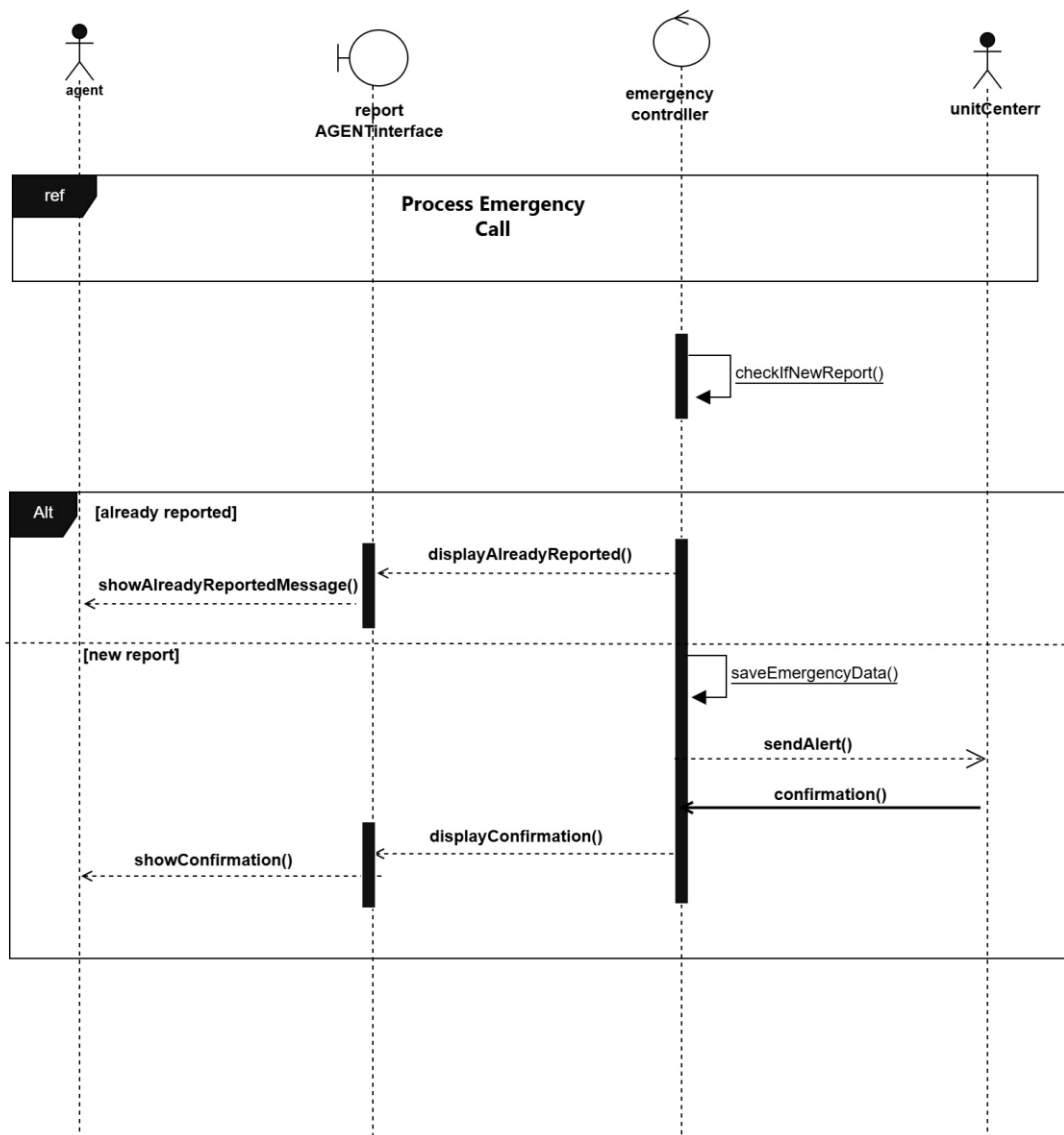
Process Emergency Call



Alert unit Center(for CITZEN)



Alert unit Center(for AGENT)



Relational Model

user		
PK	userID	integer
	name	varchar(50)
	E-mail	varchar(50)
	password	varchar(50)

Agent		
PK	employld	integer
	departmebt	varchar(50)
FK	userID	

citzen		
PK	phonenumber	integer
FK	userID	integer

EmergencyReport		
PK	reportld	integer
	timerep	integer
FK	employld	integer
FK	phonenumber	integer

EmergencyUnit		
PK	unitld	integer
	unitttype	varchar(50)
FK	Emergencyld	integer

Emergency		
PK	Emergencyld	integer
FK	reportld	integer
	time	integer
	location	varchar(50)
	type	varchar(50)

EmergencyDescription		
PK	Descriptionld	integer
	Descriptionldtext	varchar(250)
	picture	varchar(250)
	voicemail	varchar(250)
FK	Emergencyld	integer

System Architecture

We use the **MVC pattern** in our system. MVC means **Model – View – Controller**. We chose MVC because it makes the system clear and easy to change. Each part has its own job. If we change one part, the others still work.

Model

- Works with the data.
- Talks to the database.
- Stores things like citizens, agents, reports, emergencies, and units.

View

- Shows the interface to users.
- Does not make decisions.
- In our system:
 - Citizen mobile apps (reporting and tracking)
 - Agent interface

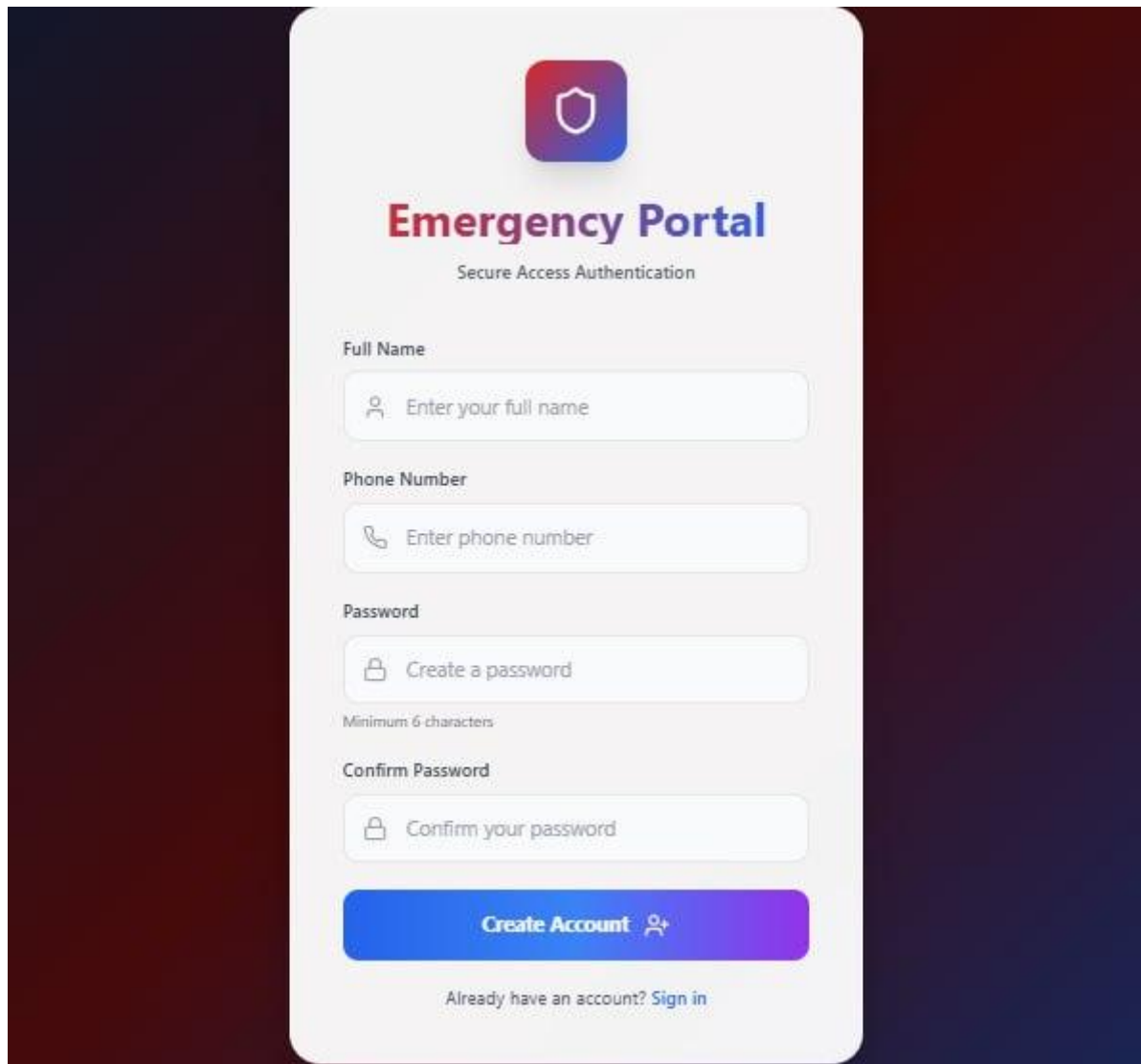
Controller

- Is the middle part.
- Connects the view and the model.
- Makes decisions.
- In our system:
 - AuthenticationController (login)
 - EmergencyController (check reports)
 - AlertController (send alerts)
 - DispatchController (send units)

Wireframe (user APP and the agent interface)

Authonitication for the app user :

Citizen can create an account and then use the app using that account any time to use the app



The wireframe shows a mobile application interface for an 'Emergency Portal'. At the top, there is a shield icon with a checkmark inside, set against a blue and purple gradient background. Below the icon, the title 'Emergency Portal' is displayed in a bold, sans-serif font, with 'Emergency' in blue and 'Portal' in purple. Underneath the title, the subtitle 'Secure Access Authentication' is written in a smaller, grey font. The main form area is a light grey rounded rectangle containing four input fields. The first field is labeled 'Full Name' and has a person icon on the left. The second field is labeled 'Phone Number' and has a phone icon on the left. The third field is labeled 'Password' and has a lock icon on the left; below it, a note says 'Minimum 6 characters'. The fourth field is labeled 'Confirm Password' and has a lock icon on the left. At the bottom of the form is a large blue button with the text 'Create Account' and a person icon. Below the button, there is a link that says 'Already have an account? Sign in'.

Emergency Portal
Secure Access Authentication

Full Name
Enter your full name

Phone Number
Enter phone number

Password
Create a password
Minimum 6 characters

Confirm Password
Confirm your password

Create Account

Already have an account? [Sign in](#)



Emergency Portal

Secure Access Authentication

Full Name



Enter your full name

Phone Number



Enter phone number

Sign In →

Don't have an account? [Create one](#)

[Report page for the app user:](#)



Emergency Report Form

Complete all required fields to submit emergency report

 **Emergency Details**

 Emergency Location *

Auto-detecting location...

 Refresh

 Date of Emergency *

jj/mm/aaaa



 Time of Emergency *

--:--



 Emergency Type *

Select emergency type



Emergency Description (Optional)

 Text

 Photo

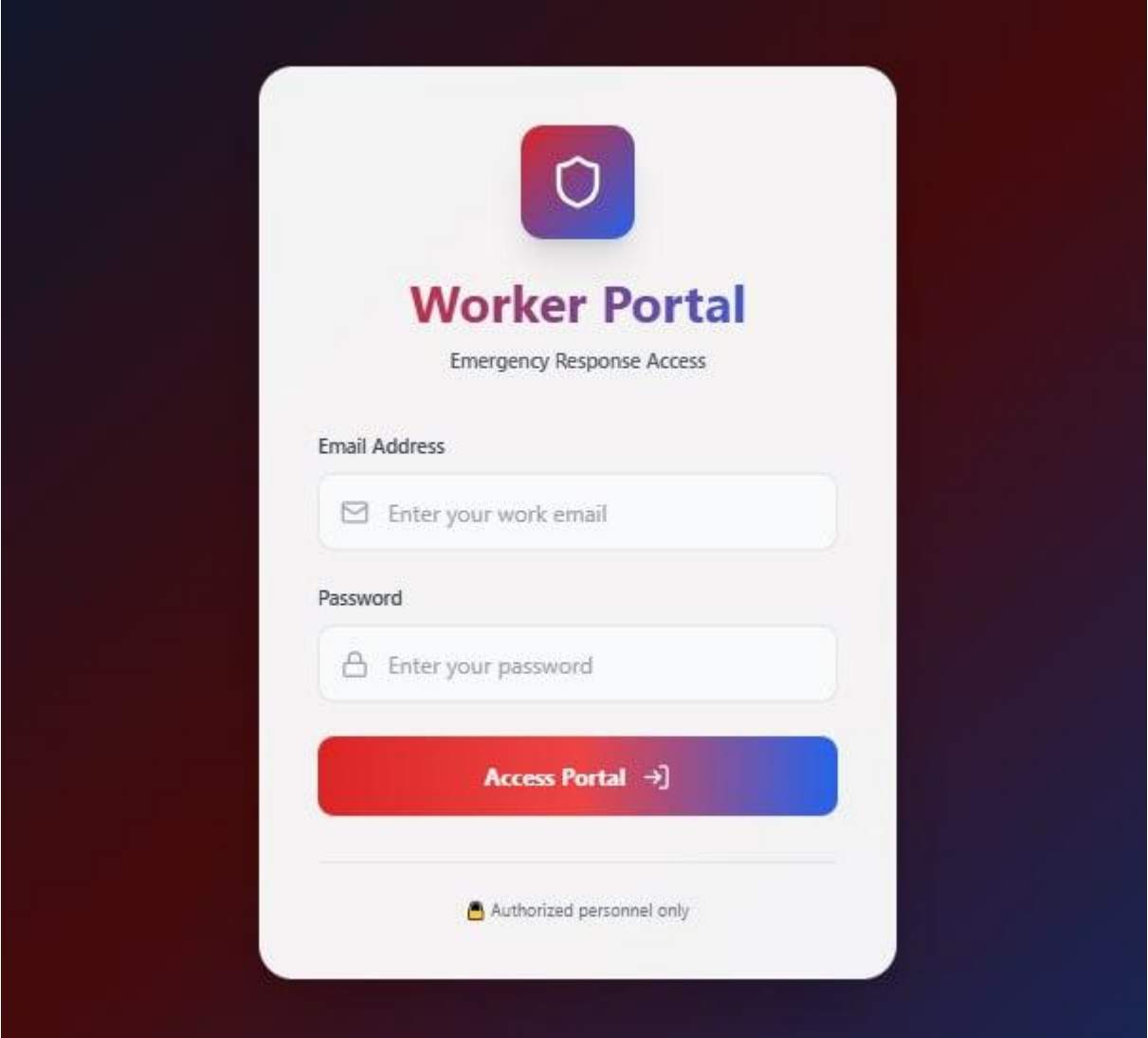
 Video

 Audio

Submit Emergency Report 

All reports are encrypted and reviewed by emergency response personnel

Authentication for the Agent:



The image shows a login interface for a 'Worker Portal'. It features a central white card on a dark blue gradient background. At the top of the card is a shield icon with a white outline. Below the icon, the text 'Worker Portal' is displayed in a large, bold, sans-serif font, with 'Worker' in red and 'Portal' in blue. Underneath this, 'Emergency Response Access' is written in a smaller, grey font. The login form consists of two input fields: 'Email Address' and 'Password'. The 'Email Address' field has a placeholder text 'Enter your work email' and a small envelope icon. The 'Password' field has a placeholder text 'Enter your password' and a small lock icon. Below these fields is a large, rounded rectangular button with a red-to-blue gradient, containing the text 'Access Portal' in white and a right-pointing arrow. At the bottom of the card, there is a small padlock icon followed by the text 'Authorized personnel only'.

Worker Portal
Emergency Response Access

Email Address

Enter your work email

Password

Enter your password

Access Portal →

Authorized personnel only

Report AGENT Interface:



Emergency Report Form

Fill report on behalf of caller

 Emergency Worker Mode

 Caller Information

Caller Name *

Caller Phone *

 Emergency Details

 Emergency Location *

 Refresh

 Date of Emergency *



 Time of Emergency *



 Emergency Type *



Submit Emergency Report 